

Preliminary Image Source Retrieval Results

M.Tech CSE | Original-image retrieval from edited-image queries

Measured local evaluation. The fusion models were trained on a bounded MagicBrush train sample and evaluated unchanged on a separate public dev split. The result below is a small improvement signal, not proof of reliable gain or novelty.

System and methods

Edited image	Frozen SSCD	Top 20 candidates	Nine local patches	Fixed score or trained MLP	Rank sources
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SSCD stays frozen. It retrieves candidates by cosine similarity. A 3 x 3 patch grid provides local evidence, and each fusion MLP combines global and local scores to rerank the top 20. Each MLP has 65 trainable parameters.

Separate MagicBrush dev results

Method	Queries / gallery	Recall@1	Recall@5	Recall@10
Fusion MLP, all-stage training	156 / 413	156/156 (100%)	1.0	1.0
Fusion MLP, first-edit training	156 / 413	156/156 (100%)	1.0	1.0
SSCD + fixed patch score	156 / 413	155/156 (99.36%)	1.0	1.0
Frozen SSCD global cosine	156 / 413	155/156 (99.36%)	1.0	1.0

Evaluation protocol

Training used 500 rows from MagicBrush train, grouped into 257 complete sources for source-separated fitting/validation/testing. The saved first-edit and multi-stage weights were loaded unchanged. Evaluation used 300 rows from the separate public dev split and retained 156 complete sessions; the final edit from each session was the query. The gallery combined 156 dev originals with 257 train-split distractor originals, for 413 images. Exact source-image hash overlap was checked; 0 overlapping sources were found. Dev examples were not used for training or tuning.

What the result says

Frozen SSCD missed rank 1 on one query: the correct source was rank 3. Fixed patch scoring moved it to rank 2. Both learned fusion models moved it to rank 1. This is one corrected case among 156 queries, so it is a lead to investigate, not a dependable improvement claim. The single-edit and multi-stage models tied exactly.

Results by edit turn

Final edit turn	Queries	SSCD R@1	Fixed patch R@1	Both MLPs R@1	All methods R@5/10
1	62	1.0	1.0	1.0	1.00 / 1.00
2	44	1.0	1.0	1.0	1.00 / 1.00
3	50	0.98	0.98	1.0	1.00 / 1.00

The one corrected query

Final edited query	Correct original	Source rank by method
		SSCD: 3 Fixed patches: 2 Single-edit MLP: 1 Multi-stage MLP: 1

The MLPs recovered this third-turn source from the same candidate set. A single query cannot establish that this behavior will repeat.

Supporting checks

On the internal 500-row train-sample holdout, all four methods reached $\text{Recall}@1/5/10 = 1.00$. Five synthetic corruption checks (two center crops, gray occlusion, blur, and JPEG compression) also scored 1.00 for every method. These checks did not distinguish the methods; the synthetic images were not AI edits.

Limitations and next step

This was a bounded prefix of MagicBrush dev, which is a development split rather than the hidden official test. It contains one separating query, no semantic hard-negative mining, and no external dataset. Avoid tuning on this dev sample again. Confirm the result on an untouched dataset or permitted test split with many more difficult edits and visually similar distractors. If multi-stage training continues to tie the single-edit model, drop the multi-stage claim.

Reproducibility

Code: `run_local_multistage.py` and `run_external_dev_eval.py`. Saved run records include split counts, checkpoint SHA-256, and protocol. Raw rankings, manifests, training logs, weights, and cached features are in `outputs/multistage_local/` and `outputs/external_dev/`. See `RESULTS.md` for exact commands. Dataset card: <https://huggingface.co/datasets/osunlp/MagicBrush>.